Introduction to Web Application Technologies

(Module 1 .a)

Behind Every Click: Browser–Server Communication



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Objective

By the end of this class, students will be able to

- **Understand** the fundamental concepts, components, and architecture of web applications.
- **Identify** the key technologies used in web application development, including HTML, CSS, JavaScript, server-side technologies, and databases.
- **Differentiate** between client-side and server-side technologies and understand their roles in web application development.
- **Discuss** the basic communication process between web browsers, web servers, application servers, and databases.
- **Recognize** common web application security challenges and the importance of secure web application development.

Why Do We Need Web Application Technologies?

Without a web application:

Customer → Call/Visit Restaurant → Place Order → Manual Recording → Food Preparation → Delivery

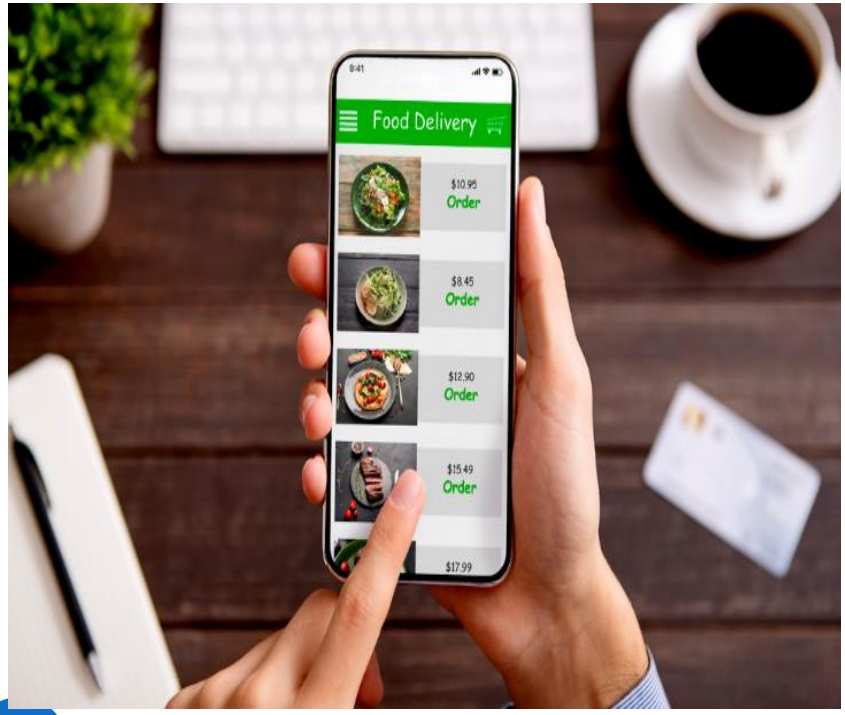




Why Do We Need Web Application Technologies?

With Web Application

Customer → Web Application → Select Food → Place Order → Automatic Processing → Food Preparation → Delivery





Why Do We Need Web Application Technologies?

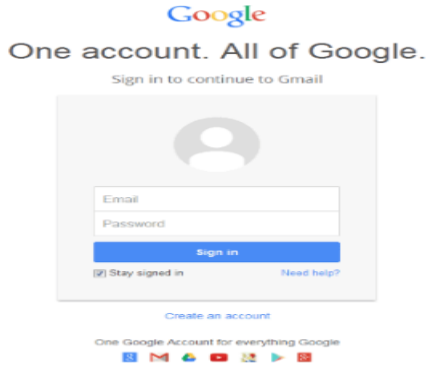
Web Application Technologies provide the technologies needed to build **interactive, accessible, and data-driven applications through a web browser.**

- **Access Anywhere** – Use applications from any device with a browser and Internet connection.
- **Connect Users & Services** – Allow users to interact with businesses, organizations, and online services.
- **Interactive Experience** – Support login, search, shopping, booking, payments, messaging, etc.
- **Manage Data** – Store, retrieve, and update information using databases.
- **Real-Time Services** – Provide continuously updated information such as orders, messages, and notifications.
- **Secure Communication** – Protect user data through authentication, HTTPS, and other security mechanisms.

Introduction to Web Application Technologies

- A **web application** is a software application accessed through a web browser.
- It uses web technologies to provide interactive services to users.
- Web applications can run on different devices such as:
  - Desktop computers
  - Laptops
  - Smartphones
  - Tablets

Examples: Online banking, e-commerce websites, email, learning management systems, social media.









What Happens When You Open a Website?

Web Browser

A **web browser** is a software application used to **access and view websites and web applications on the Internet**.

Examples

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Apple Safari

When you open Chrome and visit an online shopping website, **Chrome is the web browser**.






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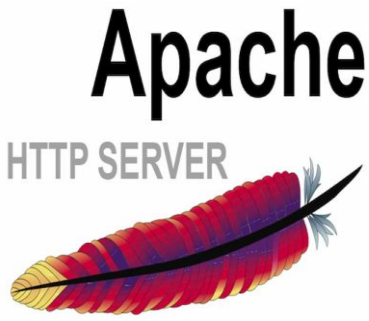
Web Server

A **web server** is a computer/system that **stores and delivers web pages and web application resources to users when requested**.

Examples of web server software:

- Apache
- Nginx
- Microsoft IIS

When you request an online shopping website, the **web server receives your request and sends the required webpage/data back to your browser**.







Example

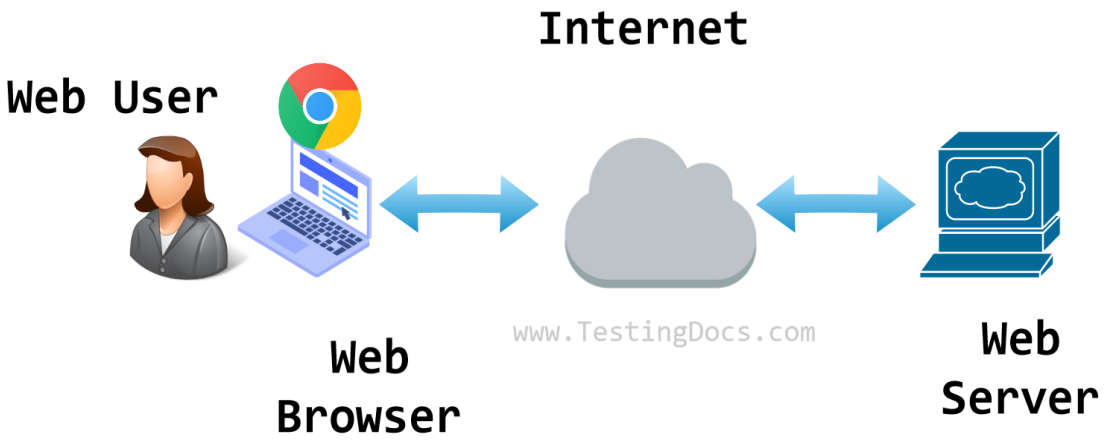
Customer → Restaurant → Customer







Example



- **Web Browser = Customer interface** → Requests and displays information.

- **Web Server = Restaurant staff** → Receives the request, processes it, and provides the requested information.

What is a Web Application?

- A web application combines **frontend, backend, and database components**.
- The **frontend** is the part users see and interact with.
- The **backend** processes requests and implements application logic.
- The **database** stores and manages application data.
- Communication between components usually takes place over the Internet using **HTTP/HTTPS**.

Example: Online Shopping

- Frontend → Product page and shopping cart
- Backend → Processes orders and payments
- Database → Stores products, customers, and orders

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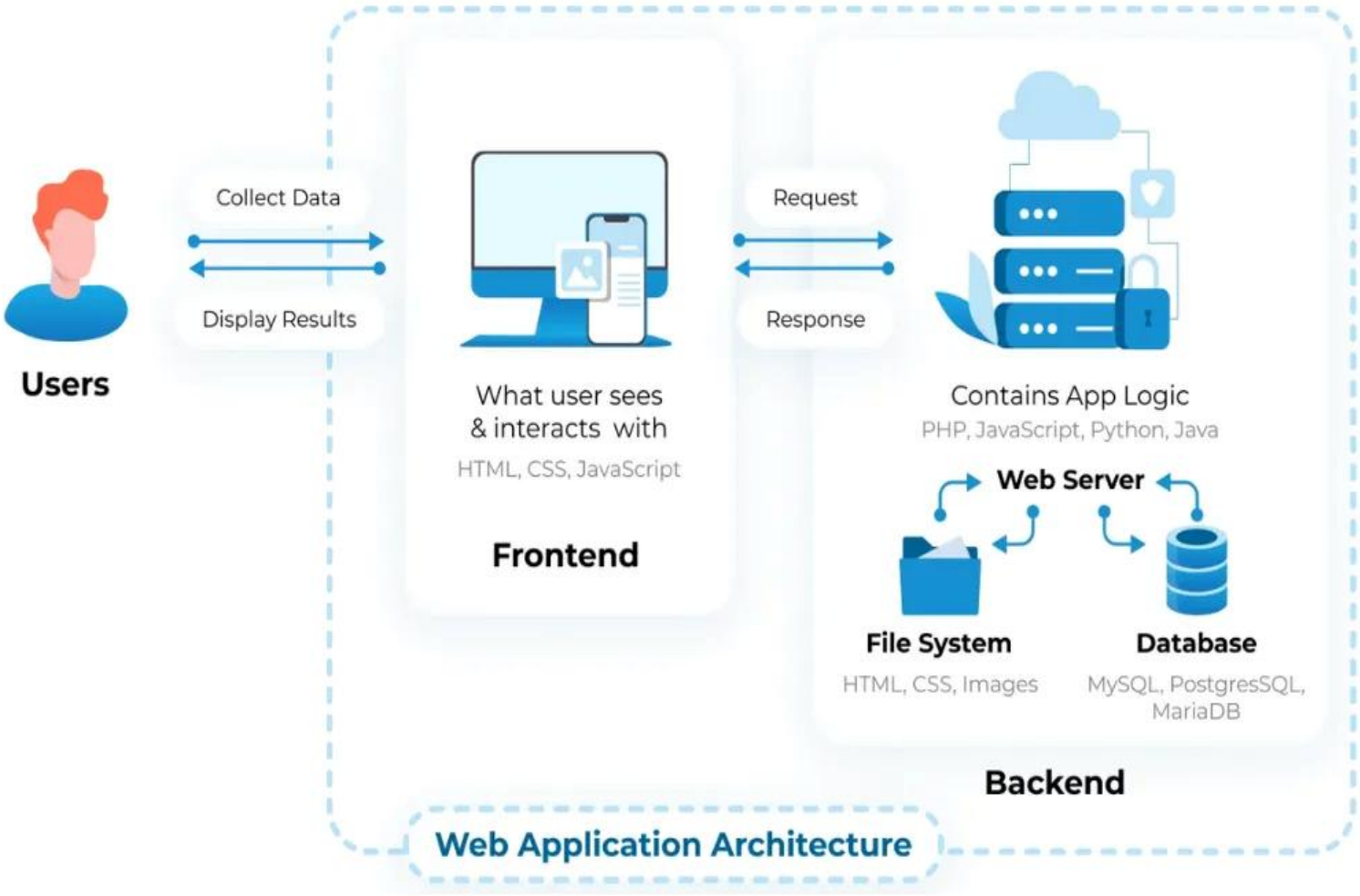
Basic Web Application Architecture

Three main components

- **Client / Frontend**
  - Runs in the user's browser.
  - Uses HTML, CSS, and JavaScript.
  - Provides the user interface.
- **Web Server / Backend**
  - Receives and processes client requests.
  - Implements business logic.
  - Communicates with databases and other services.
- **Database**
  - Stores application information.
  - Examples: User accounts, products, orders, and transactions.
- **Simple flow**

Browser → Web Server → Database → Web Server → Browser

Basic Web Application Architecture

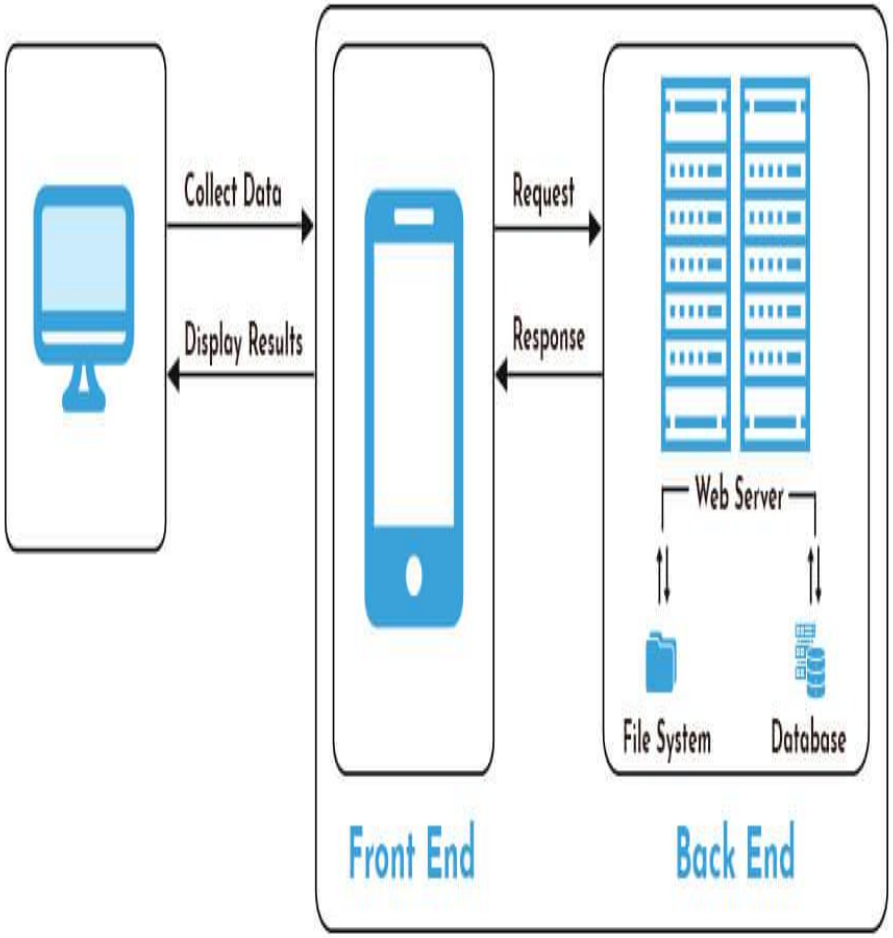


Major Web Application Technologies

Technology	Purpose
HTML (Hyper Text Markup Language)	Defines the structure of web pages
CSS (Cascading Style Sheets)	Controls presentation and layout
JavaScript (JS)	Adds interaction and dynamic behavior
HTTP/HTTPS (Hyper Text Transfer Protocol Secure)	Enables communication between client and server
PHP / Python / Java / Node.js	Backend/server-side development
MySQL / PostgreSQL / MongoDB	Data storage and management

How a Web Application Works

- **Step 1:** User enters a URL or interacts with a webpage.
- **Step 2:** Browser sends an **HTTP/HTTPS request** to the server.
- **Step 3:** Web server receives the request.
- **Step 4:** Backend application processes the request.
- **Step 5:** Backend may retrieve or update information in the database.
- **Step 6:** Server sends a response back to the browser.
- **Step 7:** Browser displays the result to the user.



Web Application Technologies and Security

Web applications handle important user information, making security essential.

Common security concerns include:

- **Authentication** – Verifying the identity of users.
- **Authorization** – Controlling what users are allowed to access.
- **Input Validation** – Checking user-supplied data.
- **Session Management** – Maintaining secure user sessions.
- **Data Protection** – Protecting sensitive information.
- **HTTPS** – Encrypting communication between browser and server.

Classroom Activity 1 : MCQ

1. What is a web application?

A. A software application accessed through a web browser

B. A computer hardware component

C. A programming language

D. A type of computer network

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A. To manufacture computer hardware

B. To build interactive and data-driven applications accessible through the web

C. To increase the physical storage capacity of a computer

D. To replace computer operating systems

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Classroom Activity 1 : MCQ

3. What is one major advantage of web applications?

A. They can only be used on one computer
B. They require a separate application for every device
C. They can be accessed from different devices using a web browser
D. They do not need any data storage

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Thank you